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Introduction

Discretization & Linearization

Euler Discretization

$$x(k) = x^c(t_0 + kT_s)$$
$$u(k) = u^c(t_0 + kT_s)$$
$$A = \mathbb{I} + T_s A^c$$
$$B = T_s B^c$$
$$C = C^c$$
$$D = D^c$$

Exact Discretization

$$A = \exp\left(A^c \cdot T_s\right)$$
$$B = \int_0^{T_s} \exp\left(A^c \cdot \tau\right) d\tau \, B^c$$
$$C = C^c$$
$$D = D^c$$

If A^c invertible:
$B = (A^c)^{-1}(A - \mathbb{I})B^c$

Linearization (Taylor)
Around $\bar{x}, \bar{u}$ (first order):

$$f(\bar{x}) + \frac{\partial f}{\partial x^\top}\Big|_{\bar{x}} \cdot (x - \bar{x})$$
$$\dot{x} = \underbrace{\frac{\partial g}{\partial x^\top}}_{A^c} \delta x + \underbrace{\frac{\partial g}{\partial u^\top}}_{B^c} \delta u$$
$$y = \underbrace{\frac{\partial h}{\partial x^\top}}_{C} \delta x + \underbrace{\frac{\partial h}{\partial u^\top}}_{D} \delta u$$

Exact Solution

$$x(t) = e^{A^c(t-t_0)}x_0 + \int_{t_0}^{t} e^{A^c(t-\tau)} B^c u(\tau)\, d\tau \qquad e^{A^c t} := \sum_{n=0}^{\infty} \frac{(A^c t)^n}{n!}$$

Controllability and Observability

The LTI system is **controllable** if the controllability matrix $\mathcal{C} = [B \quad AB \quad \dots \quad A^{n-1}B]$ has rank$(\mathcal{C}) = n$.
Controllability $\implies$ Stabilizability. A system is stabilizable if all uncontrollable modes (eigenvalues) are asymptotically stable ($|\lambda_j| < 1$).

The LTI system is **observable** if the observability matrix
$\mathcal{O} = [C^\top \quad (CA)^\top \quad \dots \quad (CA^{n-1})^\top]^\top$ has rank$(\mathcal{O}) = n$.
Observability $\implies$ Detectability. A system is detectable if all unobservable modes are asymptotically stable ($|\lambda_j| < 1$).

Stability of Nonlinear Systems

An equilibrium $\bar{x}$ is **Lyapunov stable** if $\forall\, \epsilon > 0 \;\exists\, \delta(\epsilon) > 0$ such that
$\|x(0) - \bar{x}\| < \delta(\epsilon) \Rightarrow \|x(k) - \bar{x}\| < \epsilon \;\forall k \geq 0$, and **globally asymptotically stable (GAS)** if additionally $\|x(k) - \bar{x}\| \to 0$ as $k \to \infty$ for all $x(0) \in \mathbb{R}^n$.
A **Lyapunov function** V is GAS of $\bar{x} = 0$ if: $V(0) = 0$, $V(x) > 0$ and $V(x) \to \infty$ for all $x \neq 0$, and $V(g(x)) - V(x) \leq -\alpha(x) \leq 0 \;\forall x$. If $\alpha(x)$ is only positive *semi*definite, then $\bar{x}$ is globally Lyapunov stable.
Indirect Method: Let $A = \frac{\partial g}{\partial x^T}\Big|_{x=0}$. The origin is locally asymptotically stable if $|\lambda_i| < 1 \;\forall i$, and unstable if $|\lambda_i| > 1$ for some i. If any $|\lambda_i| = 1$, no conclusion can be drawn.
Linear Systems: With $V(x) = x^T P x$ and the discrete-time Lyapunov equation
$A^T P A - P = -Q$, a unique solution $P > 0$ exists for some $Q > 0$ *iff* all eigenvalues of A lie strictly inside the unit circle. Hence, for LTI systems, GAS is both necessary and sufficient and always global.

LQR

Valid for **LTI** systems $x(k+1) = Ax(k) + Bu(k)$ with **quadratic cost function**:

$$J(x_0, U) = x_N^T P x_N + \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i \tag{1}$$

We regulate the state to the origin ($x = 0$) and we have **no** constraints. $P = P^T$ is the **terminal** weight $P \succeq 0$. $Q = Q^T$ the **state** weight $Q \succeq 0$ and $R = R^T$ is the **input** weight $R \succ 0$. $x(0)$ is the current state, $x_0, \dots, x_N$ and $u_0, \dots, u_N$ the optimization variables.

Batch approach

Results in a series of numerical values based on the initial state $x(0)$.

$$\begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} I \\ A \\ \vdots \\ \vdots \\ A^N \end{bmatrix} x(0) + \begin{bmatrix} 0 & \cdots & \cdots & 0 \\ B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \ddots & \ddots & 0 \\ A^{N-1}B & \cdots & AB & B \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ \vdots \\ u_{N-1} \end{bmatrix}$$

Or in compact notation as $X = S^x x(0) + S^u U$ and with $\bar{Q} = \text{blockdiag}(Q, \cdots, Q, P)$, $\bar{R} = \text{blockdiag}(R, \cdots, R)$. The cost function becomes:

$$J(x(0), U) = X^T \bar{Q} X + U^T \bar{R} U$$
$$= U^T H U + 2x(0)^T F U + x(0)^T S^{x^T} \bar{Q} S^x x(0)$$

with $H = (S^u)^T \bar{Q} S^u + \bar{R}$ and $F = (S^x)^T \bar{Q} S^u$. Note that $H \succ 0$. Minimizing $\nabla_U J(x(0), U) = 0$:

$$U^\star = -H^{-1} F^T x(0)$$
$$J^\star = -x(0)^T F H^{-1} F^T x(0) + x(0)^T (S^x)^T \bar{Q} S^x x(0)$$

Recursive approach

From (1) we start solving for time $N - 1$, obtaining the recursive formula:

$$u_i^\star = -(B^T P_{i+1} B + R)^{-1} B^T P_{i+1} A x(i) = F_i x_i$$
$$P_i = A^T P_{i+1} A + Q$$
$$\quad - A^T P_{i+1} B (B^T P_{i+1} B + R)^{-1} B^T P_{i+1} A$$
$$J_i^\star(x(i)) = x(i)^T P_i x(i)$$

Batch vs **Recursive**: batch adapts more easily to constraints; recursive is more efficient and robust to uncertainties.
ARE ($N \to \infty$): steady-state P_∞, $F_\infty = -(B^T P_\infty B + R)^{-1} B^T P_\infty A$

- $u^\star(k) = F_\infty x(k) \Rightarrow$ closed loop **asymptotically stable**
- $J^\star = x^T P_\infty x$ is a **Lyapunov function** of $x(k+1) = (A + BF_\infty)x(k)$

Convex Optimization

The standard form is:

$$\begin{aligned} \min_{x \in \text{dom}(f)} \quad & f(x) \\ \text{subj. to} \quad & g_i(x) \leq 0 \\ & h_i(x) = 0 \end{aligned} \tag{2}$$

The i^{th} inequality constraint $g_i(x) \leq 0$ is active at $\bar{x}$ if $g_i(\bar{x}) = 0$, otherwise it is inactive. Equality constraints are always active.

Convex Sets

A set is a **convex set** if and only if for x and $y \in \mathcal{X}$:

$$\lambda x + (1 - \lambda) y \in \mathcal{X}, \forall \lambda \in [0, 1], \forall x, y \in \mathcal{X}$$

1. **Hyperplane**: Defined as $a^T x = b$ (*affine*).
2. **Halfspace**: $a^T x \leq b$ Everything on one side of the Hyperplane.
3. **Polyhedron**: $Ax \leq b$ Intersection of a finite number of closed halfspaces.
4. **Polytope**: A bounded polyhedron.
5. **Ellipsoid**: $(x - x_c)^T A^{-1} (x - x_c) \leq 1$ with x_c the center and $A \succ 0$.
6. **Norm Balls**: $\|x - x_c\|_p \leq r$ with a radius r and a norm p

The intersection of two or more convex sets is itself convex. (Union not convex in general)

Convex Functions

A function is convex iff dom(f) is convex and $f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y)$.

1. **First-order condition**: convex iff $f(y) \geq f(x) + \nabla f(x)^T (y - x)$
2. **Second-order condition**: convex iff $\nabla^2 f(x) \succeq 0$

- **Affine**: $ax + b$ for any $a, b \in \mathbb{R}$
- **Exponential**: e^{ax} for any $a \in \mathbb{R}$
- **Powers**: x^α on $\mathbb{R}_{++}$ and $\alpha \geq 1$ or $\alpha \leq 0$
- **Vector norms**: $\|x\|_p$ for $p \geq 1$

Operations that preserve convexity: Non-negative weighted sum, Composition with affine function, Pointwise maximum or supremum, Partial minimization,...

Optimality Conditions

For a convex problem of the form (2), the **Lagrangian Function** is defined as
$L(x, \lambda, \nu) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{i=1}^{p} \nu_i h_i(x)$ The **dual function** is defined as $d(\lambda, \nu) = \inf_x L$, it is **always concave** and generates a **lower bound** $d(\lambda, \nu) \leq p^\star$. The dual problem $d^\star = \max d(\lambda, \nu)$ subject to $\lambda \geq 0$ is **always convex** even if the primal is not.

If there is at least one strictly feasible point (for convex problems), then $p^\star = d^\star$ (**Slater Condition**). Moreover the **KKT conditions** are defined as

1. **Stationarity**: $\nabla_x L = \nabla f(x^\star) + \sum \lambda_i^\star \nabla g_i(x^\star) + \sum \nu_i^\star \nabla h_i(x^\star) = 0$
2. **Primal Feasibility**: $h_i(x^\star) = 0$ and $g_i(x^\star) \leq 0$
3. **Dual Feasibility**: $\lambda^\star \geq 0$
4. **Complementarity Slackness**: $\lambda_i^\star g_i(x^\star) = 0$

For Convex with Slatter condition they are **necessary and sufficient**, but in general just **necessary**.

Sensitivity Analysis

$$\begin{aligned} \min_{x} \quad & f(x) \\ \text{subj. to} \quad & g_i(x) \leq u_i \quad i = 1 \dots m \\ & h_i(x) = v_i \quad i = 1 \dots p, \end{aligned} \qquad \begin{aligned} \max_{\nu, \lambda} \quad & d(\nu, \lambda) - u^\top \lambda - v^\top \nu \\ \text{subj. to} \quad & \lambda \geq 0 \end{aligned}$$

Assuming strong duality for the original unperturbed problem, weak duality for the perturbed case implies:

$$p^\star(u, v) \geq d^\star(\nu^\star, \lambda^\star) - u^\top \lambda^\star - v^\top \nu^\star = p^\star(0, 0) - u^\top \lambda^\star - v^\top \nu^\star$$

- **Global Sensitivity:** Evaluated by varying $\lambda_i^\star, \nu_i^\star$ and u_i, v_i.
- **Local Sensitivity:** If $p^\star(u, v)$ is differentiable at $(0, 0)$, then:

$$\lambda_i^\star = -\frac{\partial p^\star(0, 0)}{\partial u_i}, \qquad \nu_i^\star = -\frac{\partial p^\star(0, 0)}{\partial v_i}$$

  - $\lambda_i^\star$ is the sensitivity of $p^\star$ relative to the i^{th} inequality.
  - $\nu_i^\star$ is the sensitivity of $p^\star$ relative to the i^{th} equality.

Nominal MPC

Constrained Finite Time Optimal Control (CFTOC)

Solving 2 for an infinite horizon is **untractable**, that's why we solve it for a finite horizon N.
$J^\star(x(k)) = \min_u I_f(x_N) + \sum_{i=0}^{N-1} I(x_i, u_i)$, where $I_f = x_N^T P x_N$ approximates the tail of the cost and $I = x_i^T Q x_i + u_i^T R u_i$. This can be turned into a QP Problem of the form:

$$\begin{aligned} \min_{z \in \mathbb{R}^n} \quad & \frac{1}{2} z^T H z + q^T z + r \\ \text{subj. to} \quad & Gz \leq h \\ & Az = b \end{aligned} \tag{3}$$

Without Substitution

$$\begin{aligned} J^\star(x(k)) = \min_{z} \quad & [z^\top \; x(k)^\top] \begin{bmatrix} \bar{H} & 0 \\ 0 & Q \end{bmatrix} [z^\top \; x(k)^\top]^\top \\ \text{subj. to} \quad & G_{\text{in}} z \leq w_{\text{in}} + E_{\text{in}} x(k) \\ & G_{\text{eq}} z = E_{\text{eq}} x(k) \end{aligned} \qquad \bar{H} = \left[\begin{array}{cccc|ccc} Q & & & & & & \\ & \ddots & & & & & \\ & & Q & & & & \\ & & & P & & & \\ \hline & & & & R & & \\ & & & & & \ddots & \\ & & & & & & R \end{array}\right]$$

$$G_{\text{eq}} = \left[\begin{array}{ccccc|cccc} I & & & & & -B & & & \\ -A & I & & & & & -B & & \\ & -A & I & & & & & -B & \\ & & \ddots & \ddots & & & & & \ddots \\ & & & -A & I & & & & -B \end{array}\right], \; E_{\text{eq}} = \begin{bmatrix} A \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad G_{\text{in}} = \left[\begin{array}{cccc|cccc} 0 & & & & 0 & & & \\ \hline A_x & & & & 0 & & & \\ & A_x & & & & 0 & & \\ & & \ddots & & & & \ddots & \\ & & & A_x & & & & 0 \\ & & & A_f & & & & 0 \\ \hline 0 & & & & A_u & & & \\ & 0 & & & & A_u & & \\ & & \ddots & & & & \ddots & \\ & & & 0 & & & A_u & \\ & & & 0 & & & & A_u \end{array}\right] \quad w_{\text{in}} = \begin{bmatrix} b_x \\ b_x \\ b_x \\ \vdots \\ b_x \\ b_f \\ b_u \\ b_u \\ \vdots \\ b_u \\ b_u \end{bmatrix}$$

$$z = [x_1^\top \; \dots \; x_N^\top \; u_0^\top \; \dots \; u_{N-1}^\top]^\top \qquad E_{\text{in}} = [-A_x^\top \; 0 \cdots 0]^\top$$

With Substitution

Idea: Substitute the state equations $x_{i+1} = Ax_i + Bu_i \;\; x_0 = x(k)$

- **Step 1**: Rewrite the cost as:
  $J(x(k), U) = U^T H U + 2\, x(k)^T F U + x(k)^T Y x(k)$
  $J = [U^T \quad x(k)^T] \begin{bmatrix} H & F^T \\ F & Y \end{bmatrix} [U^T \quad x(k)^T]^T$
- **Step 2**: Rewrite the constraints compactly:
  $GU \leq w + Ex(k)$
  Feasible sets: $\mathcal{X} = \{x \,|\, A_x x \leq b_x\}$, $\mathcal{U} = \{u \,|\, A_u u \leq b_u\}$, $\mathcal{X}_f = \{x \,|\, A_f x \leq b_f\}$
  Reformulation:
  1. $A_x x \leq b_x \rightarrow \; A_x S^U U \leq b_x - A_x S^x x(k)$
  2. $A_u U \leq b_u$
  3. $\mathbf{0} \leq b_x - A_x x(k)$
     $\rightarrow$ Build resulting G, E and w.

Comparisons

$\forall\, \boldsymbol{x} \in \mathbb{R}^n, \; \boldsymbol{u} \in \mathbb{R}^m, horizon \to N$

	WITH substitution	WITHOUT substitution
# variables	Nm, $z = U$	$N(m+n) \quad z = [X \quad U]^T$
Pros & Cons	less variables evtl. badly conditioned extra computation	more variables sparse

∞ and 1 Norm Transformations (Reformulation in epigraph form)

$$\min_x \|x\|_\infty \text{ s.t. } Fx \leq g \qquad \Rightarrow \min_{x,t} t \text{ s.t. } Fx \leq g, \; -\mathbf{1}t \leq x \leq \mathbf{1}t$$

$$\min_x \|x\|_1 \text{ s.t. } Fx \leq g \qquad \Rightarrow \min_{x,t} \mathbf{1}^T t \text{ s.t. } Fx \leq g, \; -t \leq x \leq t$$

Invariance

Constr. Control: $x(k+1) = g(x(k), u(k)) \;\forall x, u \in \mathcal{X}, \mathcal{U}$
Goal: Find control law $u(k) = \kappa(x(k))$ s.t.

1. $\{x(k)\} \subset \mathcal{X}, \; \{u(k)\} \subset \mathcal{U} \;\; \forall k \rightarrow$ **Invariance**
2. Stability: $\lim_{k \to \infty} x(k) = 0$
3. Performance optimization
4. Maximization of set: $\{x(0) | \;\; 1 - 3 \; satisfied\}$

Definitions

1. **Invariance**: Region s.t. the **autonomous system** (e.g. $x(k+1) = g(x(k))$) satisfies the constraints $\forall t$
2. **Control Invariance**: Region in which there **exists a controller** κ (e.g. $x(k+1) = g(x(k), \kappa(x(k)))$) s.t. constraints are satisfied $\forall t$

Positively Invariant Set
$\mathcal{O}$ is positively invariant for the **autonomous** system if
$x(k) \in \mathcal{O} \Rightarrow x(k+1) \in \mathcal{O}, \;\forall k \in \{0, 1, \dots\}$
Maximal Positively Invariant Set $\mathcal{O}_\infty$
The set $\mathcal{O}_\infty \subset \mathcal{X}$ is a maximal positively invariant set w.r.t. $\mathcal{X}$ if:
1) $\mathcal{O}_\infty$ is positively invariant and 2) it contains all positive invariant sets
Pre Set Given a set S and the dynamics $x(k+1) = g(x(k))$, the pre set of S is:
$pre(S) := \{x | \; g(x) \in S\}$

e.g. Pre-set computation for linear autonomous systems
Given set S and dynamics $x(k+1) = Ax(k)$, then $pre(S) := \{x|\ Ax \in S\}$.
Assuming $S := \{x|\ Fx \le f\} \rightarrow pre(S) = \{x|\ FAx \le f\}$
Geometric condition for invariance
Set $\mathcal{O}$ is positively invariant iff: $\mathcal{O} \subseteq pre(\mathcal{O}) \Leftrightarrow pre(\mathcal{O}) \cap \mathcal{O} = \mathcal{O}$

Algorithm to compute the maximal positively invariant set

$\mathcal{O}_\infty$ for $x(k+1) = g(x(k))$ **Input:** $g, \mathcal{X}$ **Output:** $\mathcal{O}_\infty$
1. $\Omega_0 \leftarrow \mathcal{X}$
2. $\Omega_{i+1} \leftarrow \text{pre}(\Omega_i) \cap \Omega_i$
   if $\Omega_{i+1} = \Omega_i$ **then return** $\mathcal{O}_\infty = \Omega_i$ **else** repeat step 2.

Control Invariance

$\mathcal{C} \subseteq \mathcal{X}$ is control invariant if: $x(k) \in \mathcal{C} \Rightarrow \exists\ u(k) \in \mathcal{U}$ s.t. $g(x(k), u(k)) \in \mathcal{C}\ \ \forall k \in \mathbb{N}^+$
Maximal Control Invariant Set $\mathcal{C}_\infty$: $\mathcal{C}_\infty$ is maximal control invariant for $x(k+1) = g(x(k), u(k))$ if $\rightarrow$ 1) Control invariant and 2) contains all control invariant sets
Calculation of Control invariant sets:
Pre set $\rightarrow pre(S) := \{x|\ \exists u \in \mathcal{U}$ s.t $g(x, u) \in S\}$
A set $\mathcal{C}$ is control invariant iff $\mathcal{C} \subseteq pre(\mathcal{C})$. Same algorithm as above can be used.

Pre Set Computation: Controlled system

Given $x(k+1) = Ax(k) + Bu(k)$ with constraints $u(k) \in \mathcal{U} := \{u|\ Gu \le g\}$ and the set $S := \{x|Fx \le f\}$.
Then:

$$\begin{aligned} pre(S) &= \{x \mid \exists u \in \mathcal{U},\ Ax + Bu \in S\} \\ &= \{x \mid \exists u \in \mathcal{U},\ FAx + FBu \le f\} \\ &= \left\{x \,\middle|\, \exists u,\ \begin{bmatrix} FA & FB \\ 0 & G \end{bmatrix}\begin{bmatrix} x \\ u \end{bmatrix} \le \begin{bmatrix} f \\ g \end{bmatrix}\right\} \end{aligned}$$

$\rightarrow$ **projection operation**
From Control invariant set $\rightarrow$ **Control law**: Synthesizing a control law $\kappa(x)$ from a control invariant set through optimization problem below:
$\kappa(x) := \text{argmin}_u \{f(x, u)|\ g(x, u) \in \mathcal{C}, u \in \mathcal{U}\}$
NB: 1)$f(x, u)$ is any function, 2) system satisfies constraints but doesn't necessarily converge.
Control invariant set is a **powerful object**, and **if computable** it provides a direct method for determining a control law satisfying the constraints.
Issue: They are **not always computable** $\rightarrow$ MPC implicitly describes a control invariant set well representable and computable.

Practical Computation of invariant sets

Polytopes

Convex hull: for any subset $S \in \mathbb{R}^d$, co(S) of S $\rightarrow$ intersection of all convex sets containing S (*smallest convex set containing* S)
Proposition: Convex hull: convex hull of a set of points $\{v_1, \dots, v_k\} \in \mathbb{R}^d \rightarrow$ co$(\{v_1, \dots, v_k\}) := \{x|\ x = \sum_i \lambda_i v_i\ ,\ \lambda_i \ge 0\ ,\ \sum_i \lambda_i = 1\ \forall i = 1, \dots, k\}$
Minkowski-Weyl Theorem: For $P \subseteq \mathbb{R}^d$ the statements below are equivalent:
1. P is a polytope (i.e. $\exists\ A \in \mathbb{R}^{m \times d}$, $b \in \mathbb{R}^m$ s.t. $P = \{x|\ Ax \le b\}$)
2. P is finitely generated (i.e. $\exists\ \{v_i\}$ s.t. $P = \text{co}(\{v_1, \dots, v_s\})$)

Examples: Polytopes in MPC
- **Rate constraints**:
  $||x(k) - x(k+1)||_\infty \le \alpha \ \Rightarrow\ \begin{bmatrix} I & -I \\ -I & I \end{bmatrix}\begin{bmatrix} x(k) \\ x(k+1) \end{bmatrix} \le \mathbf{1}\alpha$
  $\rightarrow$ Polytopes in MPC commonly desribed by **inequalities**

Intersection of Polytopes:
Intersection $I \subseteq \mathbb{R}^n$ of sets $S \subseteq \mathbb{R}^n$, $T \subseteq \mathbb{R}^n$ $\ \rightarrow\ I = S \cap T := \{x|\ x \in S$ and $x \in T\}$
Subset Test: Is $P := \{x|\ Cx \le c\}$ contained in $Q := \{x|\ Dx \le d\}$?
$\rightarrow$ **True** if $P \subset \{x|\ D_i x \le d_i\}$ for each row D_i of D
$\rightarrow$ **support function** of the set P: $h_P(D_i) := \max_x D_i x$ subj. to $Cx \le c$ $(h_P(D_i) \le d_i\ \forall i)$

Ellipsoids

Ellipse: $E := \{x|\ (x - x_c)^T P (x - x_c) \le 1\}$ with P$\succeq 0$ sym. in $\mathbb{R}^{n \times n}$ and $x_c \in \mathbb{R}^n$
Lemma: Invariant set from Lyapunov function: $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is a Lyapunov fct. for the system $x(k+1) = g(x(k))$, then $Y := \{x|\ V(x) \le \alpha\}$ is **invariant** $\forall\ \alpha \ge 0$
Goal: find largest α s.t.:$Y_\alpha := \{x|\ x^T P x \le \alpha\} \subset \mathcal{X} := \{x|\ Fx \le f\}$
Maximum ellipsoidal invariant set: largest ellipsoid contained in a polytope, with

$$\alpha^\star = \max_\alpha \{\alpha \mid \|P^{-1/2} F_i^\top\|^2 \alpha \le f_i^2,\ \forall i\} = \min_i \frac{f_i^2}{F_i P^{-1} F_i^\top}.$$

Feasibility and Stability

Potential issues with MPC:
1. **No feasibility guarantee** $\rightarrow$MPC evtl. has no solution
2. **No stability guarantee** $\rightarrow$Trajectories evtl. do not converge to origin

Horizon	Feasibility	Stability / Cost
$N \rightarrow \infty$	Closed-loop trajectory always feasible	As in LQR: open loop = closed loop; finite cost $\Rightarrow x_k, u_k \rightarrow 0$
Finite N	Finite-horizon OCP may become infeasible	Inputs may not drive the system to the origin

Issues caused by use of $\rightarrow$**short sighted strategies**

Goal: Design finite horizon $\rightarrow$approximate infinite horizon
Idea: Introduce terminal cost and constraints to ensure feasibility and stability (mimic infinite horizon) $J^*(x(k)) = \min_U\ l_f(x_N)\ +\ \sum_{i=0}^{N-1} l(x_i, u_i)$, $x_N \in \mathcal{X}_f$
Recursive feasibility: showing the existence of a feasible control sequence at all times when starting from a feasible initial point
Stability: showing that the optimal cost function is a Lyapunov function, were **proven in lecture**: for cases: 1) Terminal constraint: $x_N = 0$, 2) Terminal constraint in some convex set $x_N \in \mathcal{X}_N$

Lyapunov Stability

- Asymptotic stability: Eqb. point $\bar{x} \in \Omega$ is asy. stable in the set $\Omega \in \mathbb{R}^n$ if it is Lyap. stable and **attractive**, i.e. $\lim_{k \to \infty} ||x(k) - \bar{x}|| = 0\ ,\ \forall x(0) \in \Omega$
  Global asy. stable if asy. stable and $\Omega = \mathbb{R}^n$
- Lyapunov fct.: Fct. $V : \mathbb{R}^n \rightarrow \mathbb{R}$ s.t.$V(0) = 0$ and $V(x) > 0\ \forall x \in \Omega \backslash \{0\}$
  $V(g(x)) - V(x) \le -\alpha(x)\ \forall x \in \Omega \backslash \{0\}$, $\alpha(x) : \mathbb{R}^n \rightarrow \mathbb{R}$
- Lyapunov (asy.) stability: For systems admitting Lyap. fcts. $V(x)$, $x = 0$ is **asymptotically stable** in Ω

Feasibility and stability guarantees proof for $\mathcal{X}_f = 0$

1. Goal: Prove recursive feasibility
   - Assume feasibility of $x(k)$ and $\{u_0^*\ ,\ u_1^*\ ,\ \dots\ ,\ u_{N-1}^*\}$ optimal control sequence at $x(k)$ with $\{x(k)\ ,\ x_1^*\ ,\ \dots\ ,\ x_N^*\}$ corresponding state trajectory
   - Apply $u(k) = u_0^*$ and let the system evolve to $x(k+1) = A\,x(k) + B\,u(k) = x_1^*$
   - At $x(k+1)$ the control sequence $\tilde{U} = \{u_1^*\ ,\ u_2^*\ , \dots ,\ u_{N-1}^*\ ,\ 0\}$ is **feasible**.
     $A\,x_N^* + B \cdot 0 = 0 \Rightarrow$Feasible set is invariant, terminal constraint is satisfied, Feasible set is invariant $\rightarrow$Recursive feasibility
2. Goal: Show $J^*(x(k+1)) < J^*(x(k))\ \forall x(k) \neq 0$
   $\tilde{U} = \{u_1^*\ ,\ u_2^*\ , \dots ,\ u_{N-1}^*\ ,\ 0\}$, $\tilde{X} = \{x_1^*\ ,\ \dots\ ,\ x_N^*\ ,\ 0\}$
   $J^*(x(k)) = l_f(x_N^*) + \sum_{i=0}^{N-1} l(x_i^*\ ,\ u_i^*)$, with $l_f(x_N^*) = 0$
   $J^*(x(k+1)) \le \tilde{J}(x(k+1)) = \sum_{i=1}^{N-1} l(x_i^*\ ,\ u_i^*) + l(x_N^*\ ,\ 0)$
   $= \sum_{i=0}^{N-1} l(x_i^*\ ,\ u_i^*) - l(x_0^*\ ,\ u_0^*) + l(x_N^*\ ,\ 0)$
   $= J^*(x(k)) - l(x(k)\ ,\ u_0^*) + l(0\ ,\ 0)$ where $l(0\ ,\ 0) = 0$
   $\Rightarrow J^*(x)$ is a Lyapunov function $\rightarrow$Asymptotic Lyapunov stability

Extension to more general terminal sets
Problem: The constraint $x_N = 0$ reduces the size of the feasible set
Goal: Make use of $\mathcal{X}_f$ to increase the region of attraction
$\rightarrow$Generalize proof to the constraint $x_N \in \mathcal{X}_f$
Def: Invariant set
$\mathcal{O} \rightarrow$ **positively invariant** for $x(k+1) = g_{cl}(x(k))$ if:
$x(0) \in \mathcal{O} \Rightarrow x(k) \in \mathcal{O},\ \forall k \in \mathbb{N}_+$

Stability of MPC - Main results

Assumptions (**sufficient** conditions that guarantee stability and feasibility of the MPC)
1. **Positive definite stage cost** (strictly pos. and only zero at the origin)
2. **Recursive Feasibility**: Terminal set is **invariant** under the local control law $\kappa_f(x_i)$:
   $x_{i+1} = A\,x_i + B\kappa_f(x_i) \in \mathcal{X}_f,\ \forall x_i \in \mathcal{X}_f$
   All state and input constraints are satisfied:
   $\mathcal{X}_f \subseteq \mathcal{X}\ ,\ \kappa_f(x_i) \subseteq \mathcal{U},\ \forall x_i \in \mathcal{X}_f$
3. **Stability**: Terminal cost is a continuous **Lyapunov function** in the terminal set $\mathcal{X}_f$ and:
   $l_f(x_{i+1}) - l_f(x_i) \le -l(x_i\ ,\ \kappa_f(x_i))$

Plan for N steps $\rightarrow$use local controller that keeps the system in a stable manner in the terminal set (condition 3 ensures stability)
Theorem (Closed-Loop System Stability):
Under those 3 assumptions, the closed-loop system under the MPC control law $u_0^*(x)$ is asymptotically stable and the set $\mathcal{X}_N$ (Region of Attraction) is positively invariant for the system $x(k+1) = A\,x(k) + B\,u_0^*(k)$
Stability of MPC - Outline of the Proof
- Assume feasibility of $x(k)$ and $\{u_0^*\ ,\ \dots\ ,\ u_{N-1}^*\}$ the optimal control sequence computed at $x(k)$ with $\{x(k)\ ,\ x_1^*\ ,\ \dots\ ,\ x_N^*\}$ the resulting state trajectory
- At $x(k+1) = x_1^*$ the sequence: $\tilde{U} = \{u_1^*\ ,\ \dots\ ,\ \kappa_f(x_N^*)\}$ is **feasible** $(x_N^* \in \mathcal{X}_f \rightarrow \kappa_f(x_N^*)$ is feasible and $A\,x_N^* + B\,\kappa_f(x_N^*) \in \mathcal{X}_f)$
  $\Rightarrow$Terminal constraint provides Recursive feasibility

Asymptotic stability of MPC - Outline of the proof
$J^*(x(k)) = \sum_{i=0}^{N-1} l(x_i^*\ ,\ u_i^*) + l_f(x_N^*)$
At $x(k+1) = x_1^*$, $\tilde{U} = \{u_1^*\ ,\ \dots\ ,\ \kappa_f(x_N^*)\}$ is feasible and sub-optimal:
$J^*(x(k+1)) \le \sum_{i=1}^{N-1} l(x_i^*\ ,\ u_i^*) + l(x_N^*\ ,\ \kappa_f(x_N^*)) + l_f(A\,x_N^* + B\,\kappa_f(x_N^*))$
$= \sum_{i=0}^{N-1} l(x_i^*\ ,\ u_i^*) - l(x_0^*\ ,\ u_0^*) + l(x_N^*\ ,\ \kappa_f(x_N^*)) + l_f(A\,x_N^* + B\,\kappa_f(x_N^*))$
$= J^*(x(k)) - l(x(k)\ ,\ u_0^*) + l_f(A\,x_N^* + B\,\kappa_f(x_N^*)) - l_f(x_N^*) + l(x_N^*\ ,\ \kappa_f(x_N^*))$
$\Rightarrow\ J^*(x(k+1)) - J^*(x(k)) \le -l(x(k)\ ,\ u_0^*)$
$J^*(x)$ **is a Lyapunov function**
$\Rightarrow$The closed-loop system under the MPC control law is asymptotically stable

Choice of terminal sets and cost - Linear system dynamics and quadratic cost

$J^*(x(k)) = \min_U\ x_N^T P x_N + \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i$ (Terminal cost)

subj.to $x_{i+1} = A\,x_i + B\,u_i, \forall i = 0\ ,\ \dots\ ,\ N-1$
$x_i \in \mathcal{X}\ ,\ u_i \in \mathcal{U}\ ,\ \forall i = 0\ ,\ \dots\ ,\ N-1$
$x_N \in \mathcal{X}_f$ (Terminal constraint)
$x_0 = x(k)$
- Formulate unconstrained LQR control law:
  $F_\infty = -(B^T P_\infty B + R)^{-1} B^T P_\infty A$
  $P_\infty \rightarrow$ solution to DARE
- Terminal weight $P = P_\infty$
- Terminal set $\mathcal{X}_f$ maximum invariant set for the cl-system $x_{k+1} = (A + BF_\infty)x_k$
  $x_{k+1} = (A + BF_\infty)x_k \in \mathcal{X}_f,\ \forall x_k \in \mathcal{X}_f$
  $\mathcal{X}_f \subseteq \mathcal{X}\ ,\ F_\infty x_k \subseteq \mathcal{U}\ \forall x_k \in \mathcal{X}_f$

1. Stage cost is positive definite
2. Terminal set is invariant by construction under the local control law $\kappa_f(x) = F_\infty x$
3. Terminal cost is continuous Lyapunov function in the terminal set $\mathcal{X}_f$ and:
   $x_{k+1}^T P_\infty x_{k+1} - x_k^T P_\infty x_k =$
   $x_k^T(-P_\infty + A^T P_\infty A + F_\infty^T B^T P_\infty A - F_\infty^T R F_\infty)x_k$
   $= -x_k^T(Q + F_\infty^T R F_\infty)x_k$

$\rightarrow$All assumptions of the feasibility and stability theorem are verified
Example:
1. Compute the optimal LQR controller and cost matrices: $F_\infty,\ P_\infty$.
2. Compute the maximal invariant set $\mathcal{X}_f$ for the closed-loop system $x_{k+1} = (A + BF_\infty)x_k$, subject to

$$\mathcal{X}_{\text{cl}} := \left\{x \,\middle|\, \begin{bmatrix} A_x \\ A_u F_\infty \end{bmatrix} x \le \begin{bmatrix} b_x \\ b_u \end{bmatrix}\right\}.$$

Summary
- Terminal constraint **sufficient** condition for feasibility and stability
- Region of attraction (ROA) without terminal constraint evtl. larger than for MPC with terminal constraint
- $\mathcal{X}_f = 0$ simplest choice $\rightarrow$small ROA for small N
- Solutions available for linear systems with quadratic cost
- Practically: Enlarge horizon N, check stability by sampling
- Larger horizon $N \rightarrow$ROA reaches maximum control invariant set

Sidenotes:
- From example: Decrease in the prediction horizon causes loss of the stability properties
- Inclusion of the terminal cost and constraint provides stability

Extension to nonlinear MPC
Given nonlinear system dynamics $x(k+1) = g(x(k)\ ,\ u(k))$
$\rightarrow$Assumptions on terminal set and constraints: NOT relying on linearity $\rightarrow$Lyapunov stability used to analyze stability of nonlinear dynamical systems
Issue: determining $\mathcal{X}_f$ and l_f can be difficult
Summary
Finite-horizon MPC may not be stable
Finite-horizon MPC may not satisfy the constraints for all time
$\rightarrow$ in contrast infinite-horizon provides stability and invariance
$\rightarrow$ Infinite-horizon faked by: forcing final state to be in invariant set for which there is an invariance-inducing controller

Practical MPC

Practical Issues

MPC: Practical issues
1. **Tracking**
   - Classic MPC formulation: **regulation of system to the origin**
   - BUT most commonly: tracking of non-zero output set points
2. **Disturbance rejection**
   Issue is that a constant disturbance causes an offset from the origin/ desired point of convergence
   $\rightarrow$Goal: remove offset s.t. system converges to desired set point
3. **Feasible set**
   Constraints restrict the set of states for which the optimization problem is feasible (MPC controller only defined in the feasible set)
   $\rightarrow$Make feasible set as large as possible

MPC for Reference tracking

Given: $x(k+1) = A\,x(k) + B\,u(k)$ under the constraint sets $\mathcal{X} = \{x|G_x x \le h_x\}$, $\mathcal{U} = \{u|G_u u \le h_u\}$
Goal: Drive the system to the desired steady state condition $(x_s\ ,\ u_s)$ that yields the desired output $z(k) = Hx(k) \rightarrow r$ as $k \rightarrow \infty$
How is the MPC problem modified in order to achieve tracking?
Target condition $\rightarrow x_s = Ax_s + Bu_s$, $z_s = Hx_s = r \Leftrightarrow \begin{bmatrix} I - A & -B \\ H & 0 \end{bmatrix}\begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}$
1. **Multiple feasible** u_s: find "cheapest" steady state
   $\min u_s^T R_s u_s$ subj. to $\begin{bmatrix} I - A & -B \\ H & 0 \end{bmatrix}\begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}$, $x_s \in \mathcal{X}_f\ ,\ u_s \in \mathcal{U}$
2. General assumption: feasible problem, BUT if **NO SOLUTION EXISTS**
   $\rightarrow$Compute reachable set point closest to r
   $\min(Hx_s - r)^T Q_s (Hx_s - r)$
   subj. to $x_s = Ax_s + Bu_s$, $x_s \in \mathcal{X}$, $u_s \in \mathcal{U}$

$$\min_U \ \|z_N - Hx_N\|^2_{P_z} + \sum_{i=0}^{N-1}\left(\|z_i - Hx_s\|^2_{Q_z} + \|u_i - u_s\|^2_R\right)$$

s.t. model and constraints, $x_0 = x(k)$.

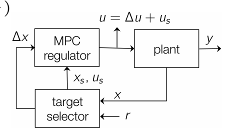
Delta formulation for tracking

Set point tracking →regulation problem with coordinate transformation

- Define deviation variables:
  $\Delta x = x - x_s$, $\Delta u = u - u_s \Rightarrow \Delta x_{k+1} = x_{k+1} - x_s$
  $= Ax_k + Bu_k - (Ax_s + Bu_s) = A\Delta x_k + B\Delta u_k$
- Constraints for deviation variables:
  $G_x x \le h_x \Rightarrow G_x \Delta x \le h_x - G_x x_s$
  $G_u u \le h_u \Rightarrow G_u \Delta u \le h_u - G_u u_s$

MPC Problem for tracking in Delta formulation:

- Initial state $\Delta x(k) = x(k) - x_s$
- Apply regulation to new system in Delta formulation:
  $\min \sum_{i=0}^{N-1} \Delta x_i^T Q \Delta x_i + \Delta u_i^T R \Delta u_i + V_f(\Delta x_N)$
  s.t. $\Delta x_0 = \Delta x(k)$
  $\Delta x_{i+1} = A\Delta x_i + B\Delta u_i$
  $G_x \Delta x_i \le h_x - G_x x_s$
  $G_u \Delta u_i \le h_u - G_u u_s$
  $\Delta x_N \in \mathcal{X}_f$



- Find optimal sequence of ΔU^*, $U^* = \Delta U^* + U_S$

MPC for tracking: Convergence

Assumption: Feasible target with $x_s \in \mathcal{X}$, $u_s \in \mathcal{U}$ with terminal weight V_f and constraint $\mathcal{X}_f$ satisfying:

- $(A + BK)x \in \mathcal{X}_f, \ \forall x \in \mathcal{X}_f$
- $V_f(x(k+1)) - V_f(x(k)) \le -l(x(k), Kx(k)), \ \forall x \in \mathcal{X}_f$

IF furthermore the target reference x_s, u_s is such that:
$x_s \oplus \mathcal{X}_f \subseteq \mathcal{X}$, $K\Delta x + u_s \in \mathcal{U}, \forall \Delta x \in \mathcal{X}_f$
→ the cl-system converges to the target reference
$x(k) \to x_s$, therefore $z(k) = Hx(k) \to r$, as $k \to \infty$

MPC for tracking: Terminal set

Issue: set of feasible targets may be reduced.
→Enlarge the set of feasible targets by scaling the terminal set for regulation

- Scale terminal set by α: $\mathcal{X}_f^{scaled} = \alpha\mathcal{X}_f$
- Thereby invariance is maintained
- Choose α s.t. state and input constraints are satisfied (scaling dependent of target)

Summary: MPC for tracking

- Set point tracking problem ⇒Regulation problem after coordinate transformation
- If cl-system stable ⇒Set point is achieved
- Difficulties:
  1. Terminal set is fct. of the target
  2. Ref. change can make the problem infeasible
  3. **Controller could show offset for unknown model error or disturbances**

MPC for reference tracking without offset

Constant disturbance acting on the system: $x(k+1) = Ax(k) + Bu(k) + B_d d$,
$y(k) = Cx(k) + C_d d$

Goal: If the system is stabilized in presence of disturbance
→it converges to set point with zero offset

Approach in constrained control:

- Model the disturbance
- Estimate state and disturbance through measurements and models
- Through disturbance estimates find control inputs that remove the offset

Augmented model

Introduce disturbance model (integral disturbance dynamics)
$x_{k+1} = Ax_k + Bu_k + B_d d_k$
$d_{k+1} = d_k$
$y_k = Cx_k + C_d d_k$
Restriction on choice of B_d, C_d →observability of the augmented model
Augmented system is observable iff:

- (A, C) is observable
- $\begin{bmatrix} A - I & B_d \\ C & C_d \end{bmatrix}$ full column rank (rank $= n_x + n_d$)
  → Maximal disturbance dimension $n_d \le n_y$

Linear state estimation

State observer for augmented problem
$$\begin{bmatrix}\hat{x}(k+1)\\ \hat{d}(k+1)\end{bmatrix} = \begin{bmatrix} A & B_d \\ 0 & I\end{bmatrix}\begin{bmatrix}\hat{x}(k)\\ \hat{d}(k)\end{bmatrix} + \begin{bmatrix}B\\0\end{bmatrix}u(k) + \begin{bmatrix}L_x\\L_d\end{bmatrix}(-y(k) + C\hat{x}(k) + C_d\hat{d}(k))$$
$\hat{(\cdot)}$→estimates of the state and disturbance, y the measured output
Error dynamics:
$$\begin{bmatrix}x(k+1) - \hat{x}(k+1)\\ d(k+1) - \hat{d}(k+1)\end{bmatrix} = \left(\begin{bmatrix} A & B_d \\ 0 & I\end{bmatrix} + \begin{bmatrix}L_x\\L_d\end{bmatrix}\begin{bmatrix}C & C_d\end{bmatrix}\right)\begin{bmatrix}x(k) - \hat{x}(k)\\ d(k) - \hat{d}(k)\end{bmatrix}$$
⇒Choose L s.t. error dynamics are asymptotically stable

Offset free tracking

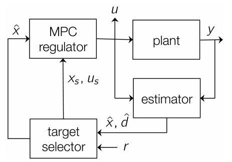
Lemma: Suppose asymptotically stable observer and $n_y = n_d$

Observer steady state satisfies: $\begin{bmatrix} A - I & B \\ C & 0\end{bmatrix}\begin{bmatrix}\hat{x}_\infty \\ u_\infty\end{bmatrix} = \begin{bmatrix} -B_d\hat{d}_\infty \\ y_\infty - C_d\hat{d}_\infty\end{bmatrix}$

⇒Observer output tracks measurement y_∞ without offset

Goal: Track constant reference r: $z(k) = Hy(k) \to r, \ k \to \infty$
New steady state condition: $x_s = Ax_s + Bu_s + B_d\hat{d}_\infty$
$$z_s = H(Cx_s + C_d\hat{d}_\infty) = r$$
General offset free tracking scheme:

1. Estimate state and disturbance $\hat{x}$, $\hat{d}$
2. Obtain (x_s, u_s) using disturbance estimate
3. Solve MPC problem using the disturbance estimate $\hat{d}$:
   $\min_U \sum_{i=0}^{N-1}(x_i - x_s)^T Q (x_i - x_s) + (u_i - u_s)^T R (u_i - u_s) + V_f(x_N - x_s)$
   subj.to $x_0 = \hat{x}(k)$, $d_0 = \hat{d}(k)$
   $x_{i+1} = Ax_i + Bu_i + B_d d_i$
   $d_{i+1} = d_i$
   $x_i \in \mathcal{X}$, $u_i \in \mathcal{U}$, $x_N - x_s \in \mathcal{X}_f$



Summary: offset-free control

Main result: IF

- $n_y = n_d$
- target steady state problem is feasible
- the cl-system converges

⇒ Target achieved without offset!

Enlarging the feasible set

Assumptions for Stability of MPC:

1. Stage cost is pos. def. (only zero at origin)
2. Terminal set is **invariant** under the local control law
   $\kappa_f(x_i) \Rightarrow x_{i+1} = Ax_i + B\kappa_f(x_i) \in \mathcal{X}_f, \ \forall x_i \in \mathcal{X}_f$
3. Terminal cost continuous Lyapunov function in $\mathcal{X}_f$ and:
   $l_f(x_{i+1}) - l_f(x_i) \le -l(x_i, \kappa_f(x_i)), \ \forall x_i \in \mathcal{X}_f$

Important: Terminal constraint →feasible set reduction
Goal: MPC without terminal constraint with guaranteed stability
Note: Feasible set without terminal constraint is NOT INVARIANT

MPC without Terminal Set

Stability is maintained when removing the terminal constraint if:
1) Initial state in sufficiently small subset of feasible set 2) Horizon N is sufficiently large
⇒Terminal state satisfies terminal constraint without explicit enforcement in the optimization
⇒Sol. of finite horizon MPC corresponding to infinite horizon MPC
Advantage: Controller defined in larger feasible set
Disadvantage: Definition of Region of attraction (ROA) or required horizon N difficult
Remarks: Larger horizon length N, ROA →maximum control inv. set

Soft constrained MPC

Input constraints usually hard vs. State/ output constraints rarely hard
→Hard state and output constraints cause complications in the controller implementation ⇒In practice: **State constraints are softened**
Objectives:

$$\left.\begin{array}{l}\text{Minimize violation duration}\\ \text{Minimize violation size}\end{array}\right\} \to \textbf{multi-objective problem}$$

Soft Constrained MPC problem setup

$$\min_{U,\epsilon} \sum_{i=0}^{N-1}\left(x_i^\top Q x_i + u_i^\top R u_i + \ell_\epsilon(\epsilon_i)\right) + x_N^\top P x_N + \ell_\epsilon(\epsilon_N)$$
s.t. $x_{i+1} = Ax_i + Bu_i$, $H_x x_i \le k_x + \epsilon_i$, $H_u u_i \le k_u$, $\epsilon_i \ge 0$,

→State constraints are **slack variables** $\epsilon_i \in \mathbb{R}^p$
→Amount of constraint violation penalized by $l_\epsilon(\epsilon_i)$
Original problem: $\min_z f(z)$ subj.to $g(z) \le 0$
"Softened" problem: $\min_{z,\epsilon} f(z) + l_\epsilon(\epsilon)$ subj.to $g(z) \le \epsilon$, $\epsilon \ge 0$
Important requirement on $l_\epsilon(\epsilon)$: If original problem has feasible solution z^*
→the softened problem should have the same solution z^*, and $\epsilon = 0$
Issue: E.g. $l_\epsilon(\epsilon) = s \cdot \epsilon^2$ doesn't satisfy this requirement for any $s > 0$
Main result: Exact Penalty function
$l_\epsilon(\epsilon) = \nu \cdot \epsilon$ satisfies the requirement for any $\nu > \lambda^* \ge 0$
(λ^* is the optimal Lagrange multiplier of teh original problem)
Extension to multiple constraints $g_j(z) \le 0$, $j = 1, \dots, r$
$l_\epsilon(\epsilon) = \nu \cdot ||\epsilon||_{1/\infty} + \epsilon^T S \epsilon$ ($\nu > ||\lambda^*||_D$ and $S \succ 0$→used to weight violations differently, $||\cdot||_D$: dual of the chosen norm)
Properties of the quadratic penalty:
→Increasing S causes **hardening of the soft constraints** (reduced size of violation BUT longer duration)
→ν large enough: satisfaction of constraints if possible (large ν: increased peak violations and decreasing duration)→**better to keep** ν **as small as possible**
Summary: Soft constraints:
→Soft constraints recover feasibility when constraints cannot be satisfied
→Allow for **tradeoff btw. violation duration vs. size**
→Good closed loop properties (standard methods of softconstrained MPC: no stability guarantees for unstable ol-systems)
→**Separation of Objectives**: Solve two optimization 1) Min. violation 2) Optimize performance (Simpler to tune but require two optimizations)

Robust MPC

Motivation: MPC relies on a model BUT noise and model inaccuracies can lead to:

- Constraint violation
- Sub-optimal behavior
- Noise prevents the system to converge to a single point

Note: Possible to incorporate some noise models in the MPC formulation (Hard to solve)

Uncertainty Models

MPC: $x(k+1) = g(x(k), u(k))$; **Reality** $x(k+1) = g(x(k), u(k), w(k); \theta)$ (Noise $\omega(k)$ & unknown model parameters θ)
Uncertain constrained system:
$x(k+1) = g(x(k), u(k), w(k); \theta)$, $x, u \in \mathcal{X}, \mathcal{U}$ $w \in \mathcal{W}$ $\theta \in \Theta$
$(\mathcal{W}, \Theta)$ assumed to be compact
Goal: Design control law $u(k) = \kappa(x(k))$ s.t.

1. Constraint satisfaction: $\{x(k)\} \subset \mathcal{X}$, $\{u(k)\} \subset \mathcal{U}$
2. Stability: Convergence to a neighborhood around the origin
3. Optimization of expected performance
4. Maximization of the set $\{x(0) |$ Conditions 1-3 satisfied $\}$

→This requires assumptions about the random values w and θ

Robust invariance

$\mathcal{O}^{\mathcal{W}}$ is robust positive invariant for the autonomous system $x(k+1) = g(x(k), w(k))$ if:
$x \in \mathcal{O}^{\mathcal{W}} \Rightarrow g(x, w) \in \mathcal{O}^{\mathcal{W}} \quad \forall w \in \mathcal{W}$
Robust Pre set:
Given Ω and $x(k+1) = g(x(k), w(k))$, the Pre set of Ω is the set of states that evolve into Ω in one time step $\forall w \in \mathcal{W}$: pre$^{\mathcal{W}}(\Omega) := \{x | \ g(x, w) \in \Omega \quad \forall w \in \mathcal{W}\}$

Computing Robust Pre-sets for linear systems

Goal: E.g. Given $g(x(k), w(k)) = Ax(k) + w(k)$ and the set $\Omega := \{x | \ Fx \le f\}$
pre$^{\mathcal{W}}(\Omega) = \{x | Ax + w \in \Omega, \ \forall w \in \mathcal{W}\} = \{x | FAx + Fw \le f, \ \forall w \in \mathcal{W}\}$
$= \{x | \ FAx \le f - \max_{w \in \mathcal{W}}\} = \{x | \ FAx \le f - h_{\mathcal{W}}(F)\}$ ($h_{\mathcal{W}}$: support function)
Geometric condition for robust invariance:
Theorem: A set $\mathcal{O}^{\mathcal{W}}$ is a robust positive invariant set iff $\mathcal{O}^{\mathcal{W}} \subseteq$ pre$^{\mathcal{W}}(\mathcal{O}^{\mathcal{W}})$
→ The algorithm to compute the robust invariant set is the same as in the nominal case, with pre(Ω) replaced by pre$^{\mathcal{W}}(\Omega)$.
Minkowski Sum: Let A, B be subsets of $\mathbb{R}^n$. The Minkowski sum is:
$A \oplus B := \{x + y | \ x \in A, \ y \in B\}$
Pontryagin Difference: Let A, B be subsets of $\mathbb{R}^n$. The Pontryagin Difference is:
$A \ominus B := \{x | \ x + e \in A \ \forall e \in B\}$

Uncertain constrained linear system

$x(k+1) = Ax(k) + Bu(k) + w(k)$ $x, u \in \mathcal{X}, \mathcal{U}$ $w \in \mathcal{W}$
Goal: Design control law $u(k) = \kappa(x(k))$ s.t.:
→Constraints are satisfied & System stabilized for all possible disturbances
Def.: $\phi_i(x_0, U, W)$: System's state at time i IF
→input $U := \{u_0, \dots, u_{N-1}\}$ applied & disturbance $W := \{w_0, \dots, w_{N-1}\}$ observed

Nominal system:
$x(k+1) = Ax(k) + Bu(k)$
$\vdots$
$x_i = A^i x_0 + \sum_{j=0}^{i-1} A^j B u_{i-1-j}$

Uncertain system:
$x(k+1) = Ax(k) + Bu(k) + w(k)$
$\vdots$
$\phi_i = x_i + \sum_{j=0}^{i-1} A^j w_{i-1-j}$

Defining cost to minimize: Cost is fct. of the disturbance seen:
$J(x_0, U, W) := \sum_{i=0}^{N-1} l(\phi_i(x_0, U, W), u_i) + l_f(\phi_N(x_0, U, W))$
Ways to eliminate the dependency on W:
→Minimize the expected value, Take the worst case, Take the nominal case (used assumption)

Robust constraint satisfaction

Idea:Formulate tighter set of constraints s.t. if nominal system meets them →then also the uncertain system does. Then do MPC on nominal system.
Goal: Ensure constraints are satisfied for the MPC sequence
$\phi_i(x_0, U, W) = \{x_i + \sum_{j=0}^{i-1} A^j w_{i-1-j} | \ W \in \mathcal{W}^i\} \subseteq \mathcal{X}$
Assumption: $\mathcal{X} = \{x | \ Fx \le f\}$
$Fx_i + F\sum_{j=0}^{i-1} A^j w_{i-1-j} \le f \quad \forall W \in \mathcal{W}^i$
$Fx_i \le f - \max_{W \in \mathcal{W}^i} F\sum_{j=0}^{i-1} A^j w_{i-1-j} = f - h_{\mathcal{W}^i}(F[A^0 \dots A^{i-1}])$
⇒**Tightening of the constraints acting on the nominal system**
Terminal state constraint
Same approach to ensure: $\phi_N(x_0, U, W) \subseteq \mathcal{X}_f$ After this:
Nominal x_i **satisfies tighter constraints→Uncertain state does too!**
$\phi_i(x_0, U, W) = \{x_i + \sum_{j=0}^{i-1} A^j w_{i-1-j} | \ W \in \mathcal{W}^i\} \subseteq \mathcal{X}$
$\Rightarrow \phi_i(x_0, U, W) \in x_i \oplus (\mathcal{W} \oplus A\mathcal{W} \oplus \dots \oplus A^{i-1}\mathcal{W}) \subseteq \mathcal{X}$
To obtain this: $x_i \in \mathcal{X} \ominus (\mathcal{W} \oplus A\mathcal{W} \oplus \dots \oplus A^{i-1}\mathcal{W}) = \mathcal{X} \ominus (\bigoplus_{j=0}^{i-1} A^j \mathcal{W})$
$\mathcal{F}_i = \bigoplus_{j=0}^{i-1} A^j \mathcal{W}$: **Disturbance reachable set** ($\mathcal{F}_{i+1} = A\mathcal{F}_i \oplus \mathcal{W}$)
Summary: Robust Open-Loop MPC
$\min_U \sum_{i=0}^{N-1} l(x_i, u_i) + l_f(x_N)$
subj.to $x_{i+1} = Ax_i + Bu_i$ (Assumption: A is stable)
$x_0 = x(k)$, $u_i \in \mathcal{U}$
$x_i \in \mathcal{X} \ominus (\mathcal{W} \oplus A\mathcal{W} \oplus \dots \oplus A^{i-1}\mathcal{W})$
$x_N \in \mathcal{X}_f \ominus (\mathcal{W} \oplus A\mathcal{W} \oplus \dots \oplus A^{N-1}\mathcal{W})$

$\mathcal{X}_f \subseteq \mathcal{X}$ robust positive invariant set for $x(k+1) = Ax(k) + w(k), \quad w(k) \in \mathcal{W}$
$\Rightarrow$**Doing nominal MPC** with **tighter state constraints!**

Closed-loop predictions

Goal: Optimize over a sequence of functions $\{u_0\,,\, \mu_1(\cdot)\,,\, \dots\,,\, \mu_{N-1}(\cdot)\}$
where $\mu_i(x_i) : \mathbb{R}^n \to \mathbb{R}^m$ is the **control policy**
For optimization assume a structure on the functions μ_i:

Pre-stabilization: $\mu_i(x_i) = Kx_i + \nu_i$
$\to$ Fixed K s.t. $A + BK$ is **stable**

Tube MPC: $\mu_i(x_i) = \nu_i + K(x_i - \bar{x}_i)$
$\to$ Fixed K s.t. $A + BK$ is **stable**
$\to$ Optimize over $\bar{x}_i$ and ν_i

Error dynamics

Main idea: $x(k+1) = Ax(k) + Bu(k) + w(k) \quad x, u \in \mathcal{X}, \mathcal{U} \quad w \in \mathcal{W}$
Separation of the control authority into two parts:

1. Portion steering the nominal system to the origin: $z(k+1) = Az(k) + B\nu(k)$
2. A portion compensating for deviations from the nominal system (**a tracking controller**): $u_i = K(x_i - z_i) + \nu_i$ (K stabilizing controller for the nominal system)

$\Rightarrow K$ is fixed offline and optimization on nominal inputs ν_i and nominal trajectory z_i
Error dynamics:
Error: $e_i = x_i - z_i$ and Feedback: $u_i = \nu_i + K(x_i - z_i)$
$\to$Error dynamics: $e_{i+1} = x_{i+1} - z_{i+1} = (A + BK)e_i + w_i$.
Note:

1. Dynamics $(A + BK)$ is **stable**
2. Set $\mathcal{W}$ is bounded

$\to$There is set ε: in which e is always contained
Goal: Bound maximum error - Uncertain state evolution: $e_{i+1} = A_K e_i + w_i, \quad e_0 = 0$
$e_i = \sum_{j=0}^{i-1} A^j w_{i-1-j}$
Set containing all possible states e_i:
$\mathcal{F}_i = \mathcal{W} \oplus \dots \oplus A^{i-1}\mathcal{W} = \bigoplus_{j=0}^{i-1} A_K^j \mathcal{W}, \quad \mathcal{F}_0 = \{0\}$
Minimum robust invariant set (mRPI):
$\mathcal{F}_\infty = \bigoplus_{j=0}^{\infty} A_K^j \mathcal{W}, \quad \mathcal{F}_0 := \{0\}$ (Insert algorithm for minimal robust invariant set)
From error bounds to robust MPC - Two variants:
Error dynamics: $e_{i+1} = (A + BK)e_i + w_i = A_K e_i + w_i \quad w_i \in \mathcal{W}$

1. Disturbance reachable sets (DRS): $e_i \in \mathcal{F}_i = \bigoplus_{j=0}^{i-1} A_K^j \mathcal{W}$, if $e_0 = 0 \to$**Constraint tightening MPC**
2. Robust positively invariant set (RPI): $e_i \in \varepsilon \Rightarrow e_{i+1} \in \varepsilon$ for any $e_0 \in \varepsilon \to e_i \in \varepsilon \to$**Tube MPC**

Robust Constraint-tightening MPC

Noisy system trajectory:
Given nominal trajectory z_i the noisy system trajectory behaves as follows:
$x_i = z_i + e_i$. Don't know the magnitude of the error BUT it will be in the set
$\mathcal{F}_i = \bigoplus_{j=0}^{i-1} A_K^j \mathcal{W}, \quad \mathcal{F}_0 := \{0\}, \quad x_i \in z_i \oplus \mathcal{F}_i = \{z_i + e_i |\, e_i \in \mathcal{F}_i\}$
Constraint tightening - Disturbance reachable sets:
Goal: $x_i\,,\, u_i \in \mathcal{X} \times \mathcal{U} \quad \forall \{w_0\,,\, \dots\,,\, w_{i-1}\} \in \mathcal{W}^i$.
To work with the nominal system $z(k+1) = Az(k) + B\nu(k)$ ensuring constraint satisfaction for the noisy sys. $x(k+1) = Ax(k) + Bu(k) + w(k)$
Necessary and suff. cond
$z_i \oplus \mathcal{F}_i \subseteq \mathcal{X} \Leftrightarrow z_i \in \mathcal{X} \ominus \mathcal{F}_i$
$u_i \in \nu_i \oplus K\mathcal{F}_i \subseteq \mathcal{U} \Leftrightarrow \nu_i \in \mathcal{U} \ominus K\mathcal{F}_i$

$$\begin{aligned} \min_{Z,V} \quad & \sum_{i=0}^{N-1} l(z_i, \nu_i) + l_f(z_N) \\ \text{s.t.} \quad & z_{i+1} = Az_i + B\nu_i, \quad z_i \in \mathcal{X} \ominus \mathcal{F}_i, \quad \nu_i \in \mathcal{U} \ominus K\mathcal{F}_i, \quad i = 0, \dots, N-1, \\ & z_N \in \mathcal{X}_f^{\text{ct}} \ominus \mathcal{F}_N, \quad z_0 = x(k), \quad \mathcal{X}_f^{\text{ct}} \text{ robust invariant.} \end{aligned}$$

Robust Invariance: If $Kz \in \mathcal{U}$ and $(A + BK)z + w \in \mathcal{X}_f^{\text{ct}}\, \forall w \in \mathcal{W}$. *The problem is recursively feasible*

Constraint-tightening MPC (Proof sketch)

- $\tilde{\nu}_i = \nu_{i+1}^* + K(\tilde{z}_i - z_{i+1}^*),\ i = 0\,,\, \dots\,,\, N-1, \quad \nu_N^* = Kz_N^*\,,\, z_{N+1}^* = A_K z_N^*$
- Recursive application:
  $\tilde{z}_i = z_{i+1}^* + A_K^i w(k),\ i = 0\,,\, \dots\,,\, N$
  $\tilde{\nu}_i = \nu_{i+1}^* + KA_K^i w(k),\ i = 0\,,\, \dots\,,\, N-1$
  By induction: $\tilde{z}_{i+1} = A\tilde{z}_i + B\tilde{\nu}_i = A(z_{i+1}^* + A_K^i w(k)) + B(\nu_{i+1}^* + K(\tilde{z} - z_{i+1}^*))$
  $= z_{i+2}^* + A_K^{i+1} w(k)$
- **Nested prediction**:
  $\tilde{z}_i \oplus \mathcal{F}_i = z_{i+1}^* + A_K^i w(k) \oplus \mathcal{F}_i \subseteq z_{i+1}^* \oplus A_K^i \mathcal{W} \oplus \mathcal{F}_i = z_{i+1}^* \oplus \mathcal{F}_{i+1} \subseteq \mathcal{X}$
- **Terminal constraint**:
  $\tilde{z}_N \oplus \mathcal{F}_N \subseteq z_{N+1}^* \oplus \mathcal{F}_{N+1} = A_K(z_N^* \oplus \mathcal{F}_N) \oplus \mathcal{W} \subseteq A_K \mathcal{X}_f \oplus \mathcal{W} \overset{\text{RPI}}{\subseteq} \mathcal{X}_f$

Robust Tube MPC

Tube MPC: Idea
Ignore noise and plan **nominal trajectory** $\to$Bound maximum error with RPI set ε:
$e_i \in \varepsilon \to e_{i+1} \in \varepsilon$

Ideally: ε minimum RPI set $\mathcal{F}_\infty = \bigoplus_{j=0}^{\infty} A^j \mathcal{W}$
Real trajectory stays "nearby" the nominal one: $x_i \in z_i \oplus \varepsilon \to$**plan to apply the controller** $u_i = K(x_i - z_i) + \nu_i$ **in the future**
$\to$Must ensure that constraints are satisfied by all possible state trajectories $z_i \oplus \varepsilon \subseteq \mathcal{X}$
Tube MPC

1. **Compute the set ε the error will remain inside (Minimum RPI set)**
   Robust positive invariant set (RPI): A set $\mathcal{O}^{\mathcal{W}}$ is RPI for the system
   $x(k+1) = g(x(k), w(k))$ if $x \in \mathcal{O}^{\mathcal{W}} \Rightarrow g(x, w) \in \mathcal{O}^{\mathcal{W}} \quad \forall w \in \mathcal{W}$
   We want the **minimum robust invariant set** (smallest set state remains in despite noise)
2. **Modify constraints on the nominal trajectory** $\{z_i\}$
   The noisy system behaves as: $x_i = z_i + e_i$ (error e_i for sure in the set ε)
   $x_i \in z_i \oplus \varepsilon = \{z_i + e_i |\, e_i \in \varepsilon\}$
   Constraint tightening
   Necessary and sufficient condition $z_i \oplus \varepsilon \subseteq \mathcal{X} \Leftrightarrow z_i \in \mathcal{X} \ominus \varepsilon$
   $u_i \in K\varepsilon \oplus \nu_i \subseteq \mathcal{U} \Leftrightarrow \nu_i \in \mathcal{U} \ominus K\varepsilon$
   Formulate as convex optimization problem

Feasible set: $\mathcal{Z}(x_0) := \{Z, V |\ z_{i+1} = Az_i + B\nu_i,\ z_i \in \mathcal{X} \ominus \varepsilon,\ \nu_i \in \mathcal{U} \ominus K\varepsilon,\ z_N \in \mathcal{X}_f,\ x_0 \in z_0 \oplus \varepsilon \quad i \in [0\,,\, \dots\,,\, N-1]\}$
Cost function: $J(Z, V) := \sum_{i=0}^{N-1} l(z_i, \nu_i) + l_f(z_N)$
Optimization: $(V^*(x_0), Z^*(x_0)) = \text{argmin}_{V,Z}\{J(Z, V) |\, (Z, V) \in \mathcal{Z}(x_0)\}$
Control law: $\mu_{tube}(x) = K(x - z_0^*(x)) + \nu_0^*(x)$

$\to$Optimization of the nominal system over **tightened state and input constraints**

Proof: Robust constraint satisfaction

- **Tube MPC assumptions**
  1. Stage cost is positive definite (only zero at the origin)
  2. Invariant terminal set for the **nominal system** with local control law $\kappa_f(z)$
     $Az + B\kappa_f(z) \in \mathcal{X}_f \quad \forall z \in \mathcal{X}_f$
     Satisfaction of all tightened state and input constraints in $\mathcal{X}_f$:
     $\mathcal{X}_f \subseteq \mathcal{X} \ominus \varepsilon,\ \kappa_f(z) \in \mathcal{U} \ominus K\varepsilon \quad \forall z \in \mathcal{X}_f$
  3. Terminal cost is continuous Lyapunov fct. in $\mathcal{X}_f$
     $l_f(Az + B\kappa_f(z)) - l_f(z) \leq -l(z\,,\, \kappa_f(z)) \quad \forall z \in \mathcal{X}_f$
- **Robust invariance of Tube MPC** (Theorem):
  $\mathcal{Z} := \{x |\, \mathcal{Z}(x) \neq 0\}$ is a **robust invariant set of**
  $\to x(k+1) = Ax(k) + B\mu_{tube}(x(k)) + w(k) \quad x, u \in \mathcal{X} \times \mathcal{U}$
  Given **optimal solution for** $x(k)$: $(\{\nu_0^*\,,\, \dots\,,\, \nu_{N-1}^*\}, \{z_0^*\,,\, \dots\,,\, z_N^*\})$
  State at next time step: $x(k+1) = Ax(k) + BK(x(k) - z_0^*) + B\nu_0^* + w$ ($x(k+1)$ may have may possible values)
  $\to$**Prove existence of feasible solution for all of them**
  By construction: $x(k+1) \in z_1^* \oplus \varepsilon \quad \forall \mathcal{W}$
  As for standard MPC the sequence:
  $(\{\nu_1^*\,,\, \dots\,,\, \nu_{N-1}^*\,,\, \kappa_f(z_N^*)\}\,,\, \{z_1^*\,,\, \dots\,,\, z_N^*\,,\, Az_N^* + B\kappa_f(z_N^*)\})$
  $\to$**feasible for all** $x(k)$

Proof: Robust stability of the cl-system
Robust stability of Tube MPC (Theorem): State $x(k)$ of the system
$x(k+1) = Ax(k) + B\mu_{tube}(x(k)) + w(k) \to$converges in the limit to set ε.
$J^*(z_0^*(x(k))) = \sum_{i=0}^{N-1} l(z_i^*, \nu_i^*) + l_f(z_N^*)$
$J^*(z_0^*(x(k+1))) \leq \sum_{i=1}^{N} l(z_i^*, \nu_i^*) + l_f(z_{N+1})$
$= J^*(z_0^*(x(k))) - l(z_0^*, \nu_0^*) - l_f(z_N^*) + l_f(z_{N+1}) + l(z_N^*, \kappa_f(z_N^*))$.
$\to \lim_{k\to\infty} J^*(z_0^*(x(k))) = 0 \Rightarrow \lim_{k\to\infty} z_0^*(x(k)) = 0$
$\to x(k)$ stays within a robust invariant set centered at $z_0^*(x(k)) : \lim_{k\to\infty} \text{dist}(x(k), \varepsilon) = 0$

Summary: Tube MPC

$$\begin{aligned} \min_{Z,V} \quad & \sum_{i=0}^{N-1} l(z_i, \nu_i) + l_f(z_N) \\ \text{s.t.} \quad & z_{i+1} = Az_i + B\nu_i, \quad z_i \in \mathcal{X} \ominus \mathcal{E}, \quad \nu_i \in \mathcal{U} \ominus K\mathcal{E}, \quad i = 0, \dots, N-1, \\ & z_N \in \mathcal{X}_f, \quad x(k) \in z_0 \oplus \mathcal{E}, \\ & A_K\mathcal{E} \oplus \mathcal{W} \subseteq \mathcal{E}, \quad \mathcal{X}_f \subseteq \mathcal{X} \ominus \mathcal{E}, \quad \mathcal{X}_f \text{ Pos. invariant.} \end{aligned}$$

Goal: Splitting inputs in two parts
1) Steering the nominal system (ν) , 2) Compensate for noise (Ke)
Finally: Optimize nominal trajectory, s.t. deviations from it within constraints

1. **Benefits**
   - Compute one constraint tightening
   - Simple optimization problem
   - Same terminal set and cost as nominal MPC
2. **Cons**
   - Sub-optimal MPC
   - Reduced feasible set when compared to nominal MPC

Summary: Robust MPC for uncertain systems

- **Constraint tightening MPC**: Disturbance reachable sets ($\mathcal{F}_k$)
  $\Rightarrow$**Nominal MPC with tightened constraints**: $\mathcal{X} \ominus \mathcal{F}_i$, initial state $z_0 = x(k)$
- **Tube-MPC**: Robust positive invariant set (ε)
  $\Rightarrow$**Nominal MPC with tightened constraints**: $\mathcal{X} \ominus \varepsilon$, initial state $z_0 \in x(k) \oplus \varepsilon$

$\to$RPI ε: increase of complexity of offline design phase
$\to$Tube MPC: **smaller feasible set & s trong stability properties**

Robustness of nominal MPC

Goal: Control noisy system $x(k+1) = Ax(k) + Bu(k) + w(k), \quad w \in \mathcal{W}$
What happens if one ignores the noise?
$\to$No certainty about success of controller (for some states it could work)
What happens to the Lyapunov fct.?
Optimal cost $J^*(x)\to$Lyapunov function for the **nominal system**
$J^*(Ax + Bu^*(x)) - J^*(x) \leq -l(x\,,\, u^*(x))$
BUT state at net point in time is:
$x(k+1) = Ax(k) + Bu^*(x(k)) + w(k)\to$still Lyapunov decrease?
Assumption: Optimal cost J^* is Lipschitz continuous
$|J^*(Ax + Bu^*(x) + w) - J^*(Ax + Bu^*(x))| \leq$
$\gamma||Ax + Bu^*(x) + w - (Ax + Bu^*(x))|| = \gamma||w||$
Lyapunov decrease bounded as:
$J^*(Ax + Bu^*(x) + w) - J^*(x)$
$= J^*(Ax + Bu^*(x) + w) - J^*(x) - J^*(Ax + Bu^*(x)) + J^*(Ax + Bu^*(x))$
$\leq -l(x\,,\, u^*(x)) + \gamma||w||$

1. Amount of decrease grows with $||x||$
2. Amount of increase upper bounded by $\max\{||w|| \,|\, w \in \mathcal{W}\}$

$\to$Movement towards the origin until: balance between size of x and w
Special case when nominal MPC is robust
Nominal model is linear, cost function $V_N(x\,,\, U)$ is linear or quadratic, sets of constraints are polytopes $\to$Optimal cost $V_N^*(x)$ is Lipschitz continuous in $\mathcal{X}_N$
Input to state stability
$\to$With the above it was shown that the system is **Input to state stable**:
Set converges to a set around zero (size determined by the size of the noise)
Summary: Nominal MPC for uncertain systems
Idea: Ignore the noise and hope the controller works!

- **Benefits**
  $\to$Simple $\to$No knowledge of $\mathcal{W}$ required $\to$Large feasible set (**BUT it may not work**)
- **Cons**
  $\to$Difficult to determine the region of attraction (set of states for which the controller will work) $\to$Works for linear systems (for nonlinear sys. only under continuity assumptions)

Implementation methods

Explicit Solution:
Active set: $I := \{1\,,\, \dots\,,\, m\}\to$set of constraint indices
$A(x) := \{j \in I :\ G_j U^*(x) - E_j x = w_j\}$
$NA(x) := \{j \in I :\ G_j U^*(x) - E_j x < w_j\}$ ($(\cdot)_j$: j^{th} row)
Critical region: $CR_A := \{x \in \mathcal{X}_N :\ A(x) = A\}$
Sequential search: Subdivide into various critical regions $CR(B_j) = \{x |\, A_j x + b_j \leq 0\}$
Online evaluation: Point location

1. Sequential search: For each j: if $A_j x + b_j \leq 0 \to x$ is in region j
   Benefit: Linear in the number of regions
2. Logarithmic search: Offline construction of search tree
   - Find hyperplane that separates regions into two equal sized sets
   - repeat for left and right sets

   Benefit: Potentially logarithmic, BUT significant offline processing

Iterative Optimization Methods

1. Unconstrained minimization
   Necessary and sufficient condition: Assume f differentiable at x^*. If f convex $\to x^*$ is a global minimizer iff $\nabla f(x^*) = 0 \Rightarrow$Unconstrained optimization methods: **iterative methods for solving** $\nabla f(x^*) = 0$
   - **Gradient descent**: $x^{i+1} = x^i - h^i \nabla f(x^i)$
     Assumption: $||\nabla f(x) - \nabla f(y)|| \leq L||x - y|| \quad \forall x, y \in \mathbb{R}^n$
     $\to$Choose constant step size: $h^i = 1/L$
   - **Newton's method**: Minimize 2^{nd} order approximation of f at current iterate x^i
     $\Rightarrow x^{i+1} = x^i - (\nabla^2 f(x^i))^{-1} \nabla f(x^i) = x^i + \Delta x_{nt}$
     Note:$\tilde{f}$ not upper bound on $f\to$**Newton step not necessarily descent**
     $\to$Use $h^i > 0$: $x^{i+1} = x^i - h^i(\nabla^2 f(x^i))^{-1} \nabla f(x^i)$
     $\to$Find $h^i > 0$ s.t. $f(x^i + h^i \Delta x_{nt}) < f(x^i)$ (e.g. using **Line search**)
2. Constrained minimization
   - **Projected Gradient Method**: $\min f(x)$ subj. to $x \in \mathbb{Q}$
     $\Rightarrow$Incorporate constraints in gradient step: $x^{i+1} = \pi_{\mathbb{Q}}(x^i - h^i \nabla f(x^i))$ with
     $\pi_{\mathbb{Q}}(y) := \text{argmin}_x \frac{1}{2}||x - y||_2^2$ s.t. $x \in \mathbb{Q}$
   - **Interior-Point Methods**: $\min f(x)$ s.t. $g_i(x) \leq 0,\ i = 1, \dots, m\,; Cx = d$
     Idea: Iteratively solve **relaxed KKT system**
     $\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla g_i(x^*) + C^T \nu^* = 0,\ Cx^* = d,$
     $g_i(x^*) + s_i^* = 0,\ \lambda_i^* g_i(x^*) = -\kappa,\ \lambda_i^*, s_i^* \geq 0$
     At each iteration: Linearization of the above

$$\begin{bmatrix} H(x,\lambda) & C^T & G(x)^T & 0 \\ C & 0 & 0 & 0 \\ G(x) & 0 & 0 & I \\ 0 & 0 & S & \Lambda \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \nu \\ \Delta \lambda \\ \Delta s \end{bmatrix} = \begin{bmatrix} -\nabla f(x) - C^T \nu - G(x)^T \lambda \\ -Cx + d \\ -(g(x) + s) \\ -S\lambda + \nu \end{bmatrix}$$

$S = \text{diag}(s_1\,,\, \dots\,,\, s_m),\ \Lambda = \text{diag}(\lambda_1\,,\, \dots\,,\, \lambda_m),$
$H(x, \lambda) = \nabla^2 f(x) + \sum_{i=1}^{m} \lambda_i \nabla^2 g_i(x),\ G(x)^T = [\nabla g_0(x) \quad \dots \quad \nabla g_m(x)]$
Vector $\nu \in \mathbb{R}^m$: generation of **multiple resulting directions** $\Delta[x\,,\, \nu\,,\, \lambda\,,\, s](\nu)$:
$\nu = 0$: **pure Newton direction**, $\nu = \kappa\mathbf{1}$: **Centering direction**, approach central path
$\Rightarrow$Linear combination via **cenetring parameter** $\sigma \in (0, 1)$ leads to **fast convergence**

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Disclaimer

This summary was created by **Andrea Bernasconi, Filippo Frigerio**, and **Mattia Dal Bò** for the lecture **Model Predictive Control (MPC)**, FS26. It was written with the help of Claude (Anthropic).

Anyone is free to further develop and publish this summary. To avoid confusion, however, it should be clearly indicated that it is not the original version.

Note: Correctness and completeness cannot be guaranteed. This summary is still quite rough in places, and may not cover all topics discussed in the lecture.

Version: 1.0

Last updated: August 30, 2026

Source & contributions:
https://github.com/mattia-dalbo/MPC-Cheatsheet-FS26.git

Found an error? Please open an issue on GitHub, or contact the authors